

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electrical and Electronics Engineering
Academic Year: 2018- 2019 (Even Semester)

Innovative Practices Description

UNIT II

Degree, Semester & Branch	: IV Semester B.E. EEE
Course Code & Title	: EE8402 Transmission and Distribution
Name of the Faculty member	: Mr. D. Karthik Prabhu
Name of the Topic	: Substation layout, Grounding methods, types of Insulators, Types of poles
Name of the Innovative Practice	: Field Visit
Date & Duration	: 18.02.2019 & 10.40 AM to 12.35 PM
ICT Tool Used	:

Description:

- Students were exposed to the different components in a substation such as transformer, bus bar arrangements, isolators, current transformer, types of insulators, Underground Cables, circuit breaker, generators and different earthing methods.
- First took the students to incoming TANGEDCO structure and explained how we interconnect with our structure using current transformer, isolators, relays, circuit breakers.
- Then Students were explained by different types of insulators, application areas of each type of insulators and Underground cables for distribution purposes.
- Students were taken to the transformer (1250 kVA) yard, where principle of operation of Transformer, constructional features was explained and students were made to clarify their doubts.
- Students were educated by how we distribute the supply to our college and also what are the necessary components for distribution.
- Finally they were explained by importance and types of earthing.
- After completion of the visit the students were educated with different components of power station, constructional features of transformers, types of insulators, UG cables and importance of earthing.

Goals (Learning Outcomes):

- Students will be able to explain about the substation layout.
- Students will be able to explain about the grounding methods.

Use of appropriate methods:

Field visit was chosen for the topic Substation and grounding methods of electrical apparatus. Because, then only students can have an exposure to how the electrical components are arranged in a substation and what are all the essential components for a substation. Also students get an exposure about how the electrical apparatus are grounded for the protection purpose.

Effective presentation

Details of the Implementation:

- After teaching the concept of substation and grounding methods students were took to the RIT-Substation (Power House).
- At the end of the visit, I asked the students to summarize the components in a substation and students were summarized and I also helped the students for the missing components.
- After completion of the field visit I uploaded the visit report in college website.

Students Strength : 56

No. of Groups : 2

Time allotted for the visit : 100 Minutes

A view of the implementation of activity:



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- The assessment of effectiveness of the activity was felt when told most of the points.
- During the visit, I understood that there should follow the sequence while mention the components in a substation and finally I insisted the components sequentially.

Reflective Critique:

- **Benefits:**

- o Students were understood clearly by visualizing the every component in a substation through the visit.
- o Students were getting motivated to go for in-plant training and Internship.

- **Challenges:**

- o Time utilization for conducting activity.

Changes required for future (if required):

- o In future, increase the time duration for the field visit.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	3	3	3	3	-	2	2	1	-	-	-	1

References:

1. V.K.Mehta, Rohit Mehta, 'Principles of power system', S. Chand & Company Ltd, New Delhi, 2013.
2. IEEE guide for Grounding.